

Colin Hanley

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EDUCATION

University of Alabama

Bachelor of Science in Mathematics and Data Science

Tuscaloosa, AL

*Aug. 2023 – May 2027**

- GPA: 3.97/4.00
- Relevant Coursework: Machine Learning, Mathematical Foundations of Data Science, Database Management Systems, Theory of Probability, Quantum Computing
- Involvement: Data Science Club, Artificial Intelligence Club

EXPERIENCE

Undergraduate Research Assistant

Department of Mathematics - Jia Zhao PhD

January 2025 - Present

Tuscaloosa, AL

- Cleaned and visualized two decades of US groundwater data
- Developed a groundwater modeling system using Multiresolution Hash Encoding and a Transformer architecture
- ASSURE grant awardee

PSIA-Certified Snowsports Instructor

Lost Trail Powder Mountain

Dec. 2021 – Jan. 2025

Sula, MT

- Designed and implemented lesson plans for beginner and intermediate snowsports students
- Assisted in the training of new snowsports instructors entering the program
- Spearheaded improvements to the lesson-assignment process, enhancing scheduling efficiency and instructor coordination

PROJECTS

College Basketball Predictive Model | *Python, Scikit-learn*

- Built a college basketball analytics platform with custom opponent-adjusted player metrics (BPM, VORP, composite ratings) across 5,000+ D1 players and 360+ teams
- Developed automated data pipelines scraping multiple sources with fuzzy entity resolution to reconcile player identities across datasets
- Created an interactive [website](#) featuring player/team rankings, transfer portal tracking, percentile filtering, and a matchup simulator

Agent-Based COVID-19 Epidemic Simulator | *Python, Mesa*

- Developed an agent-based simulation in Python to analyze COVID-19 spread, enabling scenario testing for public health interventions like vaccine strategies, masking policy, and lockdown thresholds
- Engineered features including household/workplace contact structures, age-based risk profiles, detailed immunity profiles and dynamic lockdown triggers
- Utilized Mesa for model architecture, scheduling, spatial grid, and interaction web visualization of simulation outputs

Markovian Inning Model | *Python*

- Modeled first-inning run scoring as a 24-state absorbing Markov chain, computing NRFI probabilities via fundamental matrix inversion and 50K Monte Carlo simulations with confidence intervals
- Blended 1,100+ player profiles using log-odds matchup methods with park/platoon adjustments; derived per-batter impact and sensitivity coefficients via analytic perturbation
- Automated daily pipeline: MLB API ingestion, model execution, and static site deployment with interactive visualizations via GitHub Actions

Poacher Detection in Infrared Images | *Python, Pytorch, YOLOv5*

- Fine-tuned a custom YOLOv5 model for real-time detection of human presence in LWIR UAV imagery
- Achieved 76% recall on the targeted human class after writing a script to pseudo-label synthetic data
- Wrote and presented an academic paper discussing results

TECHNICAL SKILLS

Languages: Python, R, MATLAB, C++, SQL

Libraries: Pytorch, TensorFlow, Scikit-learn, NumPy, Pandas